The Institution of Engineers (India)

(ESTABLISHED 1920, INCORPORATED BY ROYAL CHARTER 1938)

Karnataka State Centre

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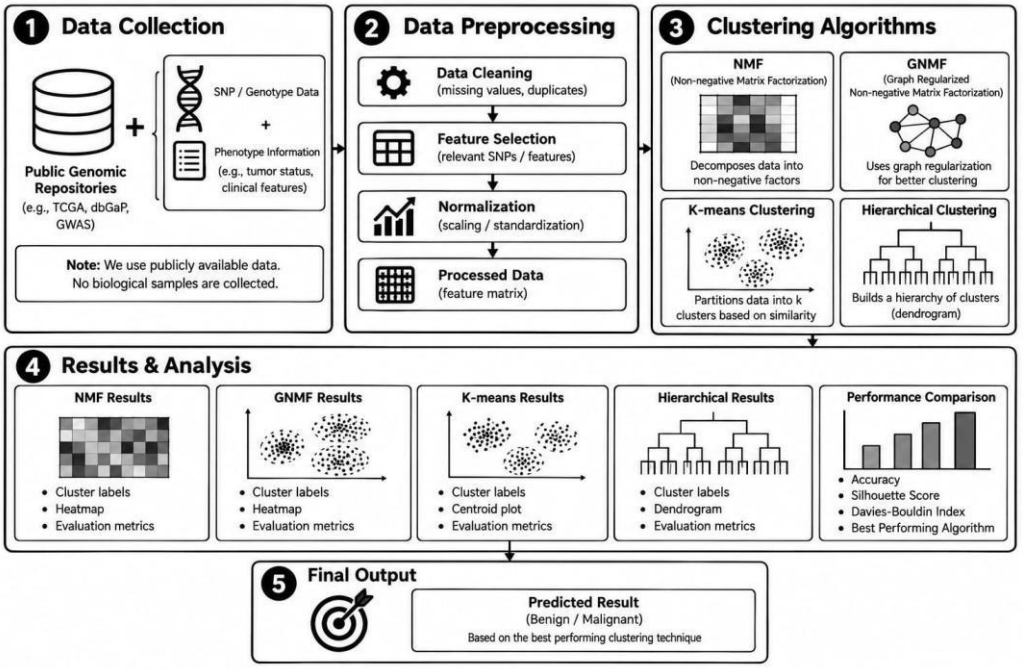
FORMAT FOR STUDENT RESEARCH PROJECT PROPOSAL

(Handwritten proposals will not be accepted, please fill all the details in this MS word file, insert images / diagrams wherever necessary. Convert to pdf file, get it approved from the project guide / Head of the Department and Head of the Institution. Hardcopy of the filled Application should be submitted to Institution of Engineers Karnataka State Centre (IEI-KSC) office.

1.	**Name of the College:** S J C INSTITUTE OF TECHNOLOGY
2.	**Project Title**: EPMIC-Breast: Predicting Breast Cancer Risk in Rural Woman Using DNA-Based Epistatic Interaction Detection and Multiple Clustering Techniques
3.	**Branch:** ARTIFICIAL INTELLIGENCE AND DATA SCIENCE
4.	**Theme (as per IEI poster):** Healthcare Management, Information Technology, Resources and Environmental Management
5.	**Name(s) of project guide(s):** **1. Name:** Mr. BADRINATH K **Email id:** badrinathk@sjcit.ac.in **Contact No.:** 9743625894 **2. Name:** Dr. SHWETHA GR **Email id:** shwethagr@sjcit.ac.in **Contact No.:** 96066 88148
6.	**Name of Team Members (Strictly not more than four students in a batch):** **Name:** SETTIVARI BHAVYA SREE **USN No.:** 1SJ23AD051 **Email id:** **bhavyasreeroyal2005@gmail.com** **Mobile No:** 7569558150 **Name:** KONDETI CHANDRA PRIYA **USN No.:** 1SJ23AD021 **Email id:** kondetichandrapriya @gmail.com

	Mobile No.: 9353969108 **Name:** MEDIMI GAGANA SREE **USN No.:** 1SJ23AD028 **Email id:** gaganasreemedimi2005@gmail.com **Mobile No.:** 7842883214 **Name:** PALLAVI B GOWDA **USN No.:** 1SJ23AD035 **Email id:** pallavibgowda3@gmail.com **Mobile No.:** 9663978936
	Team Leader of the Project: **Name:** SETTIVARI BHAVYA SREE **USN No.:** 1SJ23AD051 **Email id:** **bhavyasreeroyal2005@gmail.com** **Mobile No.:**7569558150
8.	**Processing Fee Details (Through Online Payment only):** **(processing fee of Rs. 1000/-)** **Please furnish the payment details in the format provided in the last page of the proposal.**
9.	**Date of commencement of the Project:**02/02/2026
10.	**Probable date of completion of the project:**07/11/2026
11.	**Scope / Objectives of the project:** • To collect and preprocess DNA/SNP data by converting genetic information into a suitable numerical format for efficient analysis. • To identify important SNPs and genetic patterns associated with breast cancer from high-dimensional genetic data, thereby reducing the search space and data complexity. • To apply multiple clustering techniques—NMF, GNMF, and Hierarchical K Means to discover and group similar genetic patterns and improve pattern identification. • To compare and evaluate the clustering results using suitable performance measures and visualizations to identify meaningful breast-cancer-associated patterns while reducing computational time and complexity
12.	**Methodology:** • Data Collection: Collect DNA/SNP data containing genetic variations and breast-cancer status from a suitable dataset.

- Data Preprocessing: Clean the data, handle missing values, and convert DNA/SNP information into numerical values suitable for analysis.
- Feature Preparation: Organize the SNP features and samples into a structured data matrix for clustering.
- NMF: Apply Non-negative Matrix Factorization to reduce data complexity and extract important genetic patterns.
- GNMF: Apply Graph-based NMF to consider similarities between samples and identify closely related genetic patterns



- Hierarchical K-Means: Apply Hierarchical K-Means to group samples according to their genetic similarities.
- Cluster Comparison: Compare the clusters produced by all three methods to identify common and distinct genetic patterns.
- Pattern Analysis: Analyze the resulting groups to determine patterns associated with breast-cancer samples.
- Evaluation: Evaluate and visualize the clustering results using suitable clustering measures.
- • Output: Present the identified genetic clusters and patterns through tables, graphs, and visualizations for further research.

13. **Expected Outcome of the project:**

The project is expected to provide a DNA/SNPbased framework for identifying meaningful genetic patterns associated with breast cancer using NMF, GNMF, and Hierarchical K-Means. It will simplify the analysis of complex genetic data, compare clustering results, and generate useful patterns for further research. The system can benefit researchers, bioinformatics professionals, healthcare institutions, and students by providing a practical and cost-effective analytical tool for studying breast-cancer-related genetic patterns. It can also support future development of advanced genetic risk-analysis systems, but is intended for research purposes and not direct clinical diagnosis.

	Is the project proposed relevant to the Industry / Society or Institution? **Yes / No:** **If Yes, please provide details of the Industry / institution and contact details:** **Yes.** Details of industry/institution and contact details: The project is relevant to society and healthcare research, particularly in exploring genetic patterns associated with breast cancer. It aims to develop a software-based research prototype using machine-learning and clustering techniques to analyze SNP data. The project may be relevant to cancer research institutions and healthcare research organizations. **Potential research institution:** Tata Memorial Centre – Advanced Centre for Treatment, Research and Education in Cancer (ACTREC), Kharghar, Navi Mumbai, Maharashtra, India. Official contact: mail@actrec.gov.in Telephone: +91-22-2740 5000 / 6873 5000 **Website:** https://www.actrec.gov.in/contact-us-0
15.	**Can the product or process developed in the project be taken up for filing a Patent?** **Yes / No: NO (At Present)** **Reason:** The project currently combines existing clustering algorithms such as NMF, GNMF, K-Means, and Hierarchical Clustering. A patent cannot be assumed merely because these algorithms are combined in a project. Patentability depends on whether your specific implementation has a novel, inventive, and legally patentable contribution. You can reconsider this after a patentability assessment. **Prior Literature search done?** **Yes** **[1] Breast cancer risk prediction using machine learning:** a systematic review Sadam Hussain et al. · 2024 · Frontiers in Oncology, 14, 1343627. **Relevance:** Reviews machine-learning approaches for breast cancer risk prediction using clinical, imaging, and genomic information**.** **[2] Applying whole-genome and whole-exome sequencing in breast cancer:** a review of the landscape Hetvi Ganatra et al. · 2024 · Breast Cancer, 31, 999–1009. **Relevance:** Explains genomic sequencing, SNP identification, and genetic factors associated with breast cancer. **[3] Development and validation of a reliable DNA copy-number-based machine learning algorithm for breast cancer integrative**

cluster classification: Cameron C. Young et al. · 2024 · Scientific Reports, 14, Article 11861.

Relevance: Applies machine learning to DNA copy-number data to classify breast cancer molecular clusters, making it particularly relevant to your clustering component.

[4] Artificial intelligence applications to genomic data in cancer research: a review of recent trends and emerging areas Maria Frasca et al. · 2024 · Discover Analytics, 2, Article 10.

Relevance: Reviews how AI methods analyze genomic datasets and identify patterns relevant to cancer research.

[5] Investigating the effects of artificial intelligence on the personalization of breast cancer management: a systematic study

2024 · BMC Cancer.

Relevance: Examines AI applications in breast cancer management, including the use of genomic and molecular information.

16. Budget details (break-up details should be given):

Note: IWWA will provide nominal grant support for carrying out the project by students if selected by the project selection committee.

Budget	Amount
a) Materials / Consumables (Please specify)	2,500
b) Labor (Describe)	1,500
c) Travel (Describe)	2,000
e) Miscellaneous (Please specify)	4,000
Total	10,000

17. Any other technical details (Please specify):

The proposed project, Predicting Breast Cancer Using Multiple Clusterings Through DNA, is developed using Python, Flask, MySQL, HTML, CSS, and JavaScript. It analyzes Single Nucleotide Polymorphism (SNP) data to identify genetic patterns associated with breast cancer.

The system incorporates Non-negative Matrix Factorization (NMF), Graph-regularized Non-negative Matrix Factorization (GNMF), K-Means clustering, and Hierarchical Clustering. Jaccard similarity analysis is used to compare relevant genetic patterns.

	A web-based dashboard supports dataset uploading, merging, analysis, and visualization of results. Where suitable labeled training data and validated evaluation methods are available, a classification model can estimate whether a sample resembles the affected or unaffected groups. The project aims to explore genetic data efficiently and support research into potential gene interactions. Clustering similarity alone does not establish cancer probability, diagnose breast cancer, or determine its clinical stage. The system is an academic research prototype and requires independent clinical validation before any medical use.

Name of the Project Guide:

Email id: badrinathk@sjcit.ac.in

Contact No.: 9743625894

Name of the HOD:

Email id: hodaids@sjcit.ac.in

Contact No.: 9972925814

DECLARATION

(From Project Students)

(To scan this page and enclose in the project proposal)

We, the project team hereby declare that the details enclosed in the project proposal (Title of the Project: **EPMIC-Breast: Predicting Breast Cancer Risk in Rural Woman Using DNA-Based Epistatic Interaction Detection and Multiple Clustering Techniques**, Branch: **ARTIFICIAL INTELLIGENCE AND DATA SCIENCE**, College: **S J C INSTITUTE OF TECHNOLOGY** aretrue and correct to the best of our knowledge and belief and we undertake to inform IEI of any changes therein in the project title, students name will be intimated immediately through project guide. In case any of the above information is found to be false or untrue or misleading, we are aware that we may be held liable for it. We hereby authorize sharing of the project information with this project proposal with The Institution of Engineers, Karnataka Centre. We are aware that the project team must exhibit / demonstrate the project in the IEI Bangalore centre and interact regarding project with the experts. If the student team fails to attend the evaluation in nodal centre or exhibition (if selected), the supported project amount will be returned to IEI-KSC.

We also hereby, enclose the endorsement form to IEI-KSC, Bengaluru.

Name of the students with USN No. **Signature with date**

1. **SETTIVARI BHAVYA SREE (1SJ23AD051)**
2. **KONDETI CHANDRA PRIYA (1SJ23AD021)**
3. **MEDIMI GAGANA SREE (1SJ23AD021)**
4. **PALLAVI B GOWDA (1SJ23AD035)**

(Name & Signature of Project Guide with Seal)
Email id: badrinathk@sjcit.ac.in
Contact No.: 9743625894

(Name & Signature of HOD with Seal)
Email id: hodaids@sjcit.ac.in
Contact No.: 9972925814

ENDORSEMENT

(From College, endorsement to be taken in the institution / Department Letter head)

(To scan this page and enclose in the project proposal)

This is to certify that 1) Mr. / Ms……....., 2) Mr. / Ms. ……..... 3) Mr. / Ms. ……....., 4) Mr. / Ms..... , are bonafide student(s) of Department of ....., in the degree program of our institution. If the project proposal submitted by these students under the Student Research Project Programme is selected by IEI, we will provide the requisite laboratory / Computer / infrastructure support in our college / Institution. Further we also take necessary steps to see that the project team will exhibit / demonstrate their project in the IEI Bangalore centre and in the State Level Seminar (if selected). If the student team fails to send the completed project report or fails to attend the evaluation in nodal centre or fails to attend the Exhibition/State Level Seminar, the supported project amount will be returned to IEI-KSC.

(Name & Signature of Project Guide with Seal)	**(Signature of HOD with Seal)**	**(Signature of the Principal with Seal)**
Email id:	**Email id:**	**Email id:**
Contact No.:	**Contact No.:**	**Contact No.:**

DETAILS OF PROCESSING FEES MADE THROUGH NEFT / UPI PAYMENT

(**Note:** Include this page in the hardcopy of the student research project proposal along with transfer details)

1. TITLE OF THE PROJECT	:	**EPMIC-Breast: Predicting Breast Cancer Risk in Rural Woman Using DNA-Based Epistatic Interaction Detection and Multiple Clustering Techniques**
2. NAME OF THE TEAM LEADER	:	**SETTIVARI BHAVYA SREE**
3. EMAIL ID	:	**bhavyasreeroyal2005 @gmail.com**
4. CONTACT MOBILE NO.	:	**7569558150**

PAYMENT MADE DETAILS

5. BANK REF. NO. / UTR NO. / UPI No. (12 digits)	:	668195026761
6. TRANSACTION ID	:	T261027124315348935861 7
7. NAME OF THE SENDER / ACCOUNT HOLDER and CONTACT NUMBER	:	**SETTIVARI BHAVYA SREE** **7569558150**
8. NAME OF THE BANK	:	**UNION BANK OF INDIA**
9. PROCESSING FEES	:	Rs. 1000/-
10. DATE OF PAYMENT MADE	:	27/10/2026
11. TIME	:	12:43 PM
12. MODE OF PAYMENT MADE (NEFT / UPI, PLEASE SPECIFY)	:	**Online UPI**

(Name & Signature of the team leader)

(Name & Signature of Project Guide or HOD with Seal)

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E - Mail: ieiksc2014@gmail.com ; Website : www.ieiksc.org.in

STUDENT RESEARCH PROJECT PROGRAMME (SRPP)

(Note: This page is for information about bank details of IEI to the student team and college / institution and not to include this page in the project proposal hardcopy)

BANK ACCOUNT DETAILS

Name and address of the Institution	The Institution of Engineers (India) Dr. B. R. Ambedkar Veedhi, opp. Indian Express Building, Vasanth Nagar, Bengaluru, Karnataka 560001
Account holder's name / Designation	The Institution of Engineers (India)
Bank Account No.	54044624289
Name of the bank	State Bank of India
IFSC Code	SBIN0040022
Bank Branch Address	DR AMBEDKAR VEEDHI, VISVESWARAYA TOWER BANGALORE